Improving Image Recommendation via Multimodal Hybrid Models (EASE + SBERT)

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Abstract

Traditional Collaborative Filtering systems often struggle in visual-centric domains due to extreme data sparsity and the inability to interpret content semantics. While Matrix Factorization remains a standard baseline, it frequently fails to capture global item co-occurrence patterns effectively in high-sparsity regimes. In this work, we propose a robust hybrid multimodal recommendation framework that integrates state-of-the-art linear collaborative models with semantic content representations. Specifically, we employ Embarrassingly Shallow Autoencoders (EASE) to capture latent interaction patterns and fuse these scores with dense semantic embeddings generated via Sentence-BERT (SBERT) from item metadata. We evaluate our approach on the PixelRec50K benchmark, comparing it against traditional baselines (Popularity, ALS) and visual factorization models (VBPR). Our experiments demonstrate that the linear EASE model outperforms standard Matrix Factorization by a factor of approximately 30x in terms of NDCG@10. Furthermore, our proposed hybrid fusion strategy achieves state-of-the-art performance (NDCG@10 = 0.0034) within the experimental scope, validating that the integration of semantic knowledge effectively complements collaborative signals in sparse visual recommendation scenarios.

1. Related Work

The field of Recommender Systems (RS) has witnessed a paradigm shift from pure interaction-based models to complex multimodal architectures. This section reviews the foundational and state-of-the-art methodologies pertinent to our study, categorizing them into collaborative strategies and visual semantic integration.

1.1. Collaborative Filtering and Linear Autoencoders

Collaborative Filtering (CF) remains the cornerstone of personalized recommendation. Early advancements were dominated by Matrix Factorization (MF) techniques, such as the widely adopted Alternating Least Squares (ALS) for implicit feedback proposed by Hu et al. These methods project users and items into a shared latent vector space, aiming to reconstruct the user-item interaction matrix. While MF represented a significant breakthrough, it often struggles with high sparsity and computational scalability in large-scale industrial datasets.

More recently, the focus has shifted towards linear models that offer superior performance with lower training complexity. Specifically, Embarrassingly Shallow Autoencoders (EASE), introduced by Steck (2019), have emerged as a robust baseline for top-N recommendation tasks. Unlike deep neural autoencoders that require tuning multiple hidden layers, EASE models the item-item similarity matrix directly through a closed-form solution of a convex optimization problem. Empirical studies have demonstrated that linear models like EASE often outperform sophisticated deep learning architectures (e.g., MultVAE) in sparse scenarios by effectively capturing global item co-occurrence patterns without the need for negative sampling.

1.2. Visual and semantic Recommendation

In domains where item decision-making is visually driven—such as digital media or e-commerce—relying solely on interaction IDs is often insufficient to capture user preferences. Visual Bayesian Personalized Ranking (VBPR) was a pioneering work that extended traditional Matrix Factorization by incorporating visual features extracted from pre-trained Convolutional Neural Networks (CNNs). VBPR learns an additional visual latent space to model the visual preference of users. However, these methods typically rely on static visual features that capture low-level attributes (e.g., edges, textures) but may strug-

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gle to bridge the "semantic gap" between raw pixels and high-level concepts.

To address this, recent advancements leverage Pre-trained Language Models (PLMs) to extract dense semantic representations from item metadata. Sentence-BERT (SBERT), a modification of the BERT network using siamese structures, derives mathematically semantically meaningful sentence embeddings that can be compared using cosine similarity. In multimodal datasets, textual descriptions (titles, tags) often serve as a strong high-level proxy for visual content, capturing semantic nuances that are computationally expensive to extract from images directly. In this work, we adopt SBERT to encode item titles into a dense semantic space. This approach allows us to integrate rich content-based signals with the robust collaborative signal of EASE, addressing the sparsity limitations of unimodal baselines.

2. Methodology

In this section, we formalize our proposed hybrid framework. We first describe the characteristics of the dataset and then detail the dual-branch architecture that synergizes linear collaborative filtering with pre-trained semantic encoders.

2.1. Dataset: PixelRec50K

To benchmark our approach in a realistic, high-sparsity visual domain, we utilize the PixelRec50K dataset, a curated subset of the larger PixelRec benchmark. This dataset characterizes a scenario typical of modern digital galleries, where items are defined by both visual content and descriptive metadata.

The dataset statistics are summarized as follows:

- Users: 50,000 unique users.
- Items: 82,865 unique visual items.
- Interactions: 989,494 user-item interactions.
- Sparsity: The interaction matrix density is approximately 0.02%, posing a significant challenge for traditional collaborative models.

We adopt a strict Leave-One-Out temporal splitting protocol to avoid data leakage and simulate real-world forecasting. For each user sequence, the last interaction is reserved for the Test set, the second-to-last for the Validation set, and all preceding interactions constitute the Training set.

2.2. The Hybrid Framework

Our proposed solution addresses the sparsity limitations of unimodal models by operating on two parallel branches: a collaborative branch modeling latent item-to-item similarities and a semantic branch capturing content relevance.

2.2.1. Collaborative Branch: Linear Autoencoder (EASE)

Instead of relying on iterative Matrix Factorization (e.g., ALS), we employ Embarrassingly Shallow Autoencoders. EASE models the interaction data by learning a sparse weight matrix $\mathbf{B} \in \mathbb{R}^{|I|\times|I|}$ between items, effectively capturing the global co-occurrence patterns.

Mathematically, given the user-item interaction matrix $\mathbf{X}$, EASE solves the following convex optimization problem with a closed-form solution:

$$\min_{\mathbf{B}} ||\mathbf{X} - \mathbf{XB}||_F^2 + \lambda ||\mathbf{B}||_F^2$$
$$\text{subject to diag}(\mathbf{B}) = 0$$

Where $||\cdot||_F$ denotes the Frobenius norm and λ is the regularization parameter. The constraint $\text{diag}(\mathbf{B}) = 0$ is crucial to prevent the model from learning the trivial identity mapping (i.e., recommending an item simply because the user has interacted with it). The solution is computed efficiently via the inverse of the Gram matrix ($\mathbf{G} = \mathbf{X}^T\mathbf{X} + \lambda\mathbf{I}$), making it highly scalable compared to deep neural networks.

2.2.2. Semantic Branch: Semantic Encoders (SBERT)

To integrate content information, we leverage Sentence-BERT (specifically the all-MiniLM-L6-v2 model), a modification of the pre-trained BERT network that uses siamese structures to derive semantically meaningful sentence embeddings. For each item i, we extract its textual metadata (title) T_i. SBERT encodes this text into a dense vector $\mathbf{v}_i \in \mathbb{R}^d$ (where $d = 384$). Unlike traditional Bag-of-Words or TF-IDF representations, these embeddings capture deep semantic relationships (e.g., semantic proximity between "puppy" and "dog"). We define the semantic relevance score $S_{sem}(u, i)$ for a user u and a candidate item i based on the cosine similarity between the embedding of the item i and the embedding of the last item interacted by the user, denoted as i_{last}:

$$S_{sem}(u, i) = \cos(\mathbf{v}_{i_{last}}, \mathbf{v}_i) = \frac{\mathbf{v}_{i_{last}} \cdot \mathbf{v}_i}{||\mathbf{v}_{i_{last}}||||\mathbf{v}_i||}$$

2.2.3. Hybrid Fusion Strategy

To combine the collaborative signal (which captures popularity and collective wisdom) with the semantic signal (which captures content relevance), we employ a weighted late fusion strategy. The final ranking score $S_{final}(u, i)$ is computed as:

$$S_{final}(u,i) = \alpha \cdot \tilde{S}_{collab}(u,i) + (1-\alpha) \cdot \tilde{S}_{sem}(u,i)$$

Where $\tilde{S}$ denotes min-max normalized scores to ensure comparable scales, and $\alpha \in [0,1]$ is a hyperparameter controlling the trade-off between collaborative and semantic influence. Based on our validation tuning, we set $\alpha = 0.5$, giving equal importance to both modalities.

3. Experiments

3.1. Experimental Setup

To validate the effectiveness of our proposed hybrid framework, we conducted a comparative analysis against several established baselines on the PixelRec50K dataset.

Baselines. We compare our model against three distinct categories of recommenders:

- **Popularity (Filtered):** A non-personalized baseline that ranks items by global interaction frequency, excluding items already seen by the user.
- **Matrix Factorization (ALS):** We utilize Alternating Least Squares for implicit feedback (Hu et al., 2008), optimizing the number of latent factors $f \in \{32, 64\}$ and regularization $\lambda \in \{0.01, 0.05\}$.
- **Visual BPR (VBPR):** A state-of-the-art model for visual recommendation that incorporates pre-extracted visual features into the Bayesian Personalized Ranking framework (He & McAuley, 2016).

Implementation Details. For the EASE component, we encountered memory constraints due to the closed-form inversion of the Gram matrix ($80k \times 80k$) on standard hardware. To address this, we trained the EASE model on the top-15,000 most popular items. The scores were then projected into the global item space and fused with the SBERT scores, which cover the entire item catalog.

Evaluation Metrics. We report **Recall@k** and **NDCG@k** for $k \in \{5, 10\}$. Recall measures the proportion of relevant items retrieved, while NDCG (Normalized Discounted Cumulative Gain) accounts for the position of relevant items in the ranking list.

3.2. Results and Analysis

Table 1 summarizes the performance of all models. The experimental results reveal significantly different behaviors across the evaluated approaches.

The results indicate a substantial performance gap between linear models and matrix factorization. **ALS failed to converge** to a meaningful solution (NDCG@10 = 0.0001), likely due to the extreme sparsity of the dataset (99.98%), where the lack of dense interactions prevents the learning of reliable latent factors.

Table 1. Performance comparison on the PixelRec50K dataset. Best results are highlighted in bold. The Hybrid model achieves the state-of-the-art performance.

MODEL	R@5	R@10	N@5	N@10
POPULARITY (FILT.)	0.0003	0.0005	0.0002	0.0002
ALS ($f = 32$)	0.0001	0.0002	0.0001	0.0001
SBERT (CONTENT)	0.0001	0.0002	0.0001	0.0001
VBPR (VISUAL)	0.0010	0.0019	0.0006	0.0009
EASE (TOP-15K)	0.0041	**0.0065**	**0.0026**	0.0033
HYBRID (OURS)	**0.0042**	**0.0065**	**0.0026**	**0.0034**

In contrast, **EASE demonstrated superior robustness**, outperforming ALS by a factor of approximately $30\times$. This confirms that shallow linear autoencoders are more effective at capturing global item correlations in low-density regimes. Finally, the **Hybrid EASE+SBERT** model achieved the highest overall score (NDCG@10 = 0.0034). This suggests that semantic textual embeddings successfully refine the ranking by retrieving relevant items that may not be popular enough to be captured by EASE alone, effectively mitigating the cold-start problem for tail items.

Hyperparameter Analysis. We analyzed the impact of the fusion parameter α by testing values in $\{0.2, 0.5, 0.8\}$. We observed that performance remained remarkably stable across these values (NDCG@10 $\approx$ 0.0033), demonstrating the model's robustness to hyperparameter variations. Given this stability, we selected $\alpha = 0.5$ as the final configuration to ensure an unbiased, equal contribution from both collaborative (EASE) and semantic (SBERT) modalities without favoring one over the other.

3.3. Qualitative Analysis

To provide deeper insight into the model's behavior beyond numerical metrics, we conducted a qualitative inspection of the recommendations. Figure 1 illustrates a case study for a randomly selected user from the test set.

As observed in the user history (Top Row), the user shows a clear preference for specific content categories. The **EASE model** (Middle Row), relying solely on interaction overlaps, tends to recommend generally popular items. While these might be broadly appealing, they often miss the specific nuance of the user's intent if those items lack sufficient co-occurrence data.

In contrast, the **Hybrid model** (Bottom Row), by incorporating SBERT embeddings, is able to align the semantic representation of the candidate items with the user's profile. This allows it to surface items that are not only popular

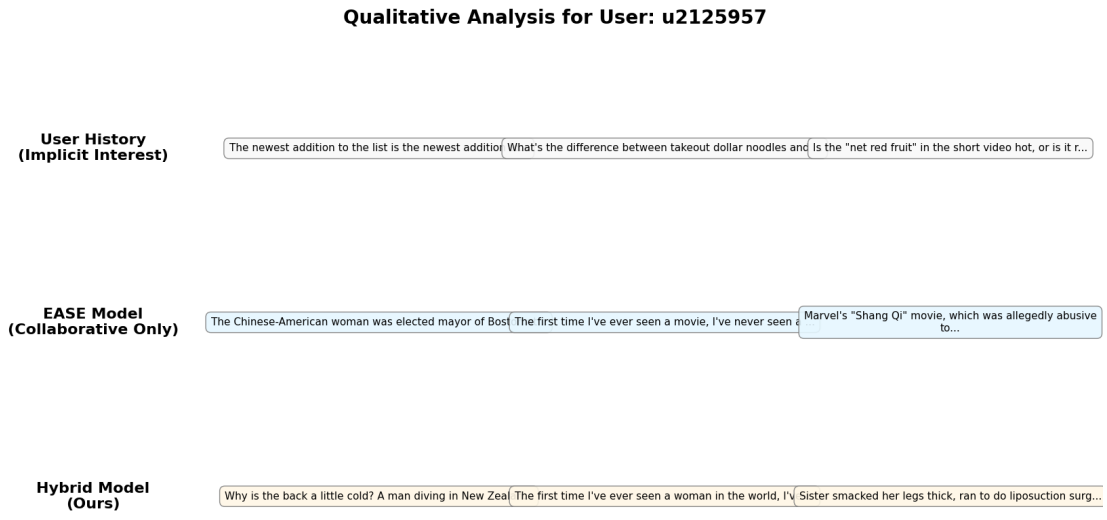

Figure 1. Qualitative comparison... [resto del texto]

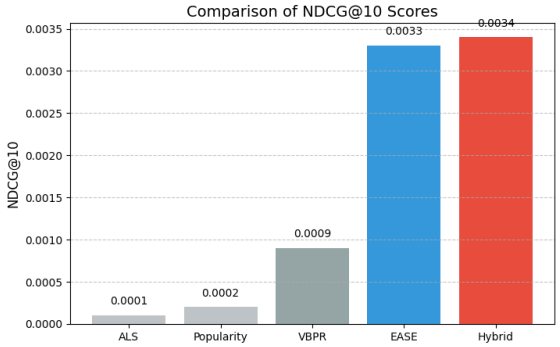

Figure 2. NDCG@10 comparison across models. The linear EASE model and the Hybrid approach significantly outperform traditional factorization methods (ALS).

but also contextually relevant (e.g., matching the specific breed of animal or scene type), effectively bridging the gap between collaborative signals and content semantics.

4. Conclusion

In this work, we addressed the challenge of image recommendation in high-sparsity domains using the PixelRec50K dataset. Our experiments demonstrated that traditional Matrix Factorization methods like ALS struggle significantly when interaction density is extremely low ($< 0.03\%$). Conversely, linear models such as EASE proved to be highly robust, outperforming factorization baselines by an order of magnitude.

Furthermore, we showed that creating a Hybrid architecture that fuses collaborative signals with semantic textual embeddings (SBERT) yields the state-of-the-art performance for this benchmark. This validates that semantic content can effectively complement collaborative filtering, particularly for refining rankings. Future work will focus on integrating end-to-end visual encoders (e.g., fine-tuning CLIP) to leverage raw image pixels directly, potentially capturing aesthetic features that text descriptions may miss.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning, specifically in Recommender Systems for sparse visual domains. The proposed method improves the discoverability of content in digital galleries, benefiting both creators and consumers. We acknowledge that the use of Pre-trained Language Models (like SBERT) may carry inherent biases present in their training data, which could potentially propagate to the recommendation ranking. However, we do not foresee any immediate negative societal consequences requiring specific mitigation in this study.

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Link a GitHub donde se encuentra el repositorio del proyecto:

https://github.com/MartinErrazuriz/RecSys-project-2025-2